# A Local All-Order Petz Theorem for the Total-Correlation Heart Problem

July 10, 2026

**Abstract**

We prove the complete Hessian comparison, at every product reference, for the Petz-overlap strengthening of Total-Correlation Heart. The result holds for every order $0 < \alpha < 1$, every number of bits, heterogeneous pure-state channels, and every tangent that preserves the one-bit marginals. Sending $\alpha \uparrow 1$ gives the corresponding local relative-entropy (TC–Heart) comparison. This is a genuinely local theorem: it does not prove the conjectured finite-displacement Petz or relative-entropy inequality.

## 1 Statement

For coordinate $i$, let a binary pure-state channel send $x \in \{0,1\}$ to the rank-one projector $P_{i,x}$. Write

$$s_i = \frac{1}{2}\|P_{i,1} - P_{i,0}\|_1 \in [0,1].$$

Thus the two signal vectors have overlap $\sqrt{1-s_i^2}$, and $s_i$ is also the reveal probability of the matched erasure channel. Let

$$q = \bigotimes_{i=1}^{n} q_i, \qquad q_i(1) = r_i, \qquad v_i = r_i(1-r_i),$$

initially with $0 < r_i < 1$ and $0 < s_i < 1$. Define the standardized coordinate

$$F_i(x_i) = \frac{x_i - r_i}{\sqrt{v_i}}.$$

The products $F_T = \prod_{i\in T} F_i$ form an orthonormal basis of $L_2(q)$. Consider a linear probability path

$$p_t = q(1+tf), \qquad f = \sum_{T\subseteq[n]} \widehat{f}(T)F_T, \qquad \mathbb{E}_q f = 0, \tag{1}$$

for all sufficiently small $|t|$. It preserves every one-bit marginal iff

$$\widehat{f}(\varnothing) = 0, \qquad \widehat{f}(\{i\}) = 0 \quad (1 \le i \le n). \tag{2}$$

Let $\Phi$ be the product pure-state channel and put

$$\rho_t = \Phi(p_t), \qquad \sigma = \Phi(q).$$

If $S$ reveals coordinate $i$ independently with probability $s_i$, define the quantum and erasure Petz overlaps

$$Q_\alpha(t) = \operatorname{Tr} \rho_t^\alpha \sigma^{1-\alpha},$$
$$C_\alpha(t) = \mathbb{E}_S \sum_z p_{t,S}(z)^\alpha q_S(z)^{1-\alpha}, \qquad 0 < \alpha < 1.$$

**Theorem 1** (Local all-order Petz comparison)**.** *For the path* (1)*, even without the marginal-preserving restriction* (2)*,*

$$Q_\alpha(0) = C_\alpha(0) = 1, \qquad Q'_\alpha(0) = C'_\alpha(0) = 0,$$

*and*

$$-Q''_\alpha(0) \le \alpha(1-\alpha) \sum_{T \neq \varnothing} \widehat{f}(T)^2 \prod_{i \in T} s_i = -C''_\alpha(0). \tag{3}$$

*In particular, for every tangent to the fixed-marginal (total-correlation) manifold, the conjectured Petz gap $Q_\alpha - C_\alpha$ has nonnegative Hessian at the product point.*

The endpoint cases in the parameters follow by continuity. We emphasize that (3) is a second-order assertion at $t = 0$; it does not assert convexity of $Q_\alpha(t) - C_\alpha(t)$ along the whole ray.

## 2 The Petz Hessian and a tensorizing upper metric

We first isolate the only operator calculation used in the proof.

**Lemma 2** (Petz Hessian kernel)**.** *Let $A > 0$ satisfy $\operatorname{Tr} A = 1$ and have eigenvalues $\lambda_a$; let $H = H^*$ with $\operatorname{Tr} H = 0$, and set*

$$\phi_\alpha(t) = \operatorname{Tr}(A + tH)^\alpha A^{1-\alpha}.$$

*Then $\phi_\alpha(0) = 1$, $\phi'_\alpha(0) = 0$, and*

$$-\phi''_\alpha(0) = \sum_{a,b} k_\alpha(\lambda_a, \lambda_b) |H_{ab}|^2, \tag{4}$$

*where*

$$k_\alpha(x,y) = \frac{(x^\alpha - y^\alpha)(x^{1-\alpha} - y^{1-\alpha})}{(x-y)^2}, \qquad k_\alpha(x,x) = \frac{\alpha(1-\alpha)}{x}.$$

*Moreover,*

$$k_\alpha(x,y) \le \frac{\alpha(1-\alpha)}{\sqrt{xy}}. \tag{5}$$

*Consequently, if*

$$\|H\|^2_{A,\mathcal{R}} := \sum_{a,b} \frac{|H_{ab}|^2}{\sqrt{\lambda_a \lambda_b}}, \tag{6}$$

*then*

$$-\phi''_\alpha(0) \le \alpha(1-\alpha) \|H\|^2_{A,\mathcal{R}}. \tag{7}$$

*Proof.* The standard second divided-difference formula for $x^\alpha$, followed by pairing the $(a,b)$ and $(b,a)$ terms, gives (4). For completeness, the diagonal term is

$$-\lambda_a^{1-\alpha}\frac{d^2}{dx^2}x^\alpha\Big|_{x=\lambda_a}|H_{aa}|^2=\frac{\alpha(1-\alpha)}{\lambda_a}|H_{aa}|^2,$$

and the paired off-diagonal terms simplify to the displayed divided-product kernel in (4).

To prove (5), use homogeneity and write $x=ge^u$, $y=ge^{-u}$, where $g=\sqrt{xy}$ and, by symmetry, $u\ge 0$. Then

$$k_\alpha(x,y)=\frac{1}{g}\frac{\sinh(\alpha u)\sinh((1-\alpha)u)}{\sinh^2 u}.$$

Convexity of sinh on $[0,\infty)$ and $\sinh 0=0$ give $\sinh(cu)\le c\sinh u$ for $0\le c\le 1$. This proves (5), and (7) follows from (4). □

The metric (6) is useful here because it tensorizes exactly. If $A=\bigotimes_i A_i$ and $H_T=\bigotimes_i H_{T,i}$, then

$$\langle H_T,H_U\rangle_{A,\mathcal{R}}=\prod_i\langle H_{T,i},H_{U,i}\rangle_{A_i,\mathcal{R}}. \tag{8}$$

## 3 The one-qubit estimate

Fix one coordinate and suppress its subscript. Put

$$\sigma=(1-r)P_0+rP_1,\qquad X=P_1-P_0,\qquad v=r(1-r).$$

**Lemma 3** (Local qubit factor)**.** *In an eigenbasis of $\sigma$, define*

$$R=\sum_{a,b}\frac{|X_{ab}|^2}{\sqrt{\lambda_a\lambda_b}}.$$

*Then*

$$vR\le s. \tag{9}$$

*Also*

$$\langle\sigma,X\rangle_{\sigma,\mathcal{R}}=0,\qquad \|\sigma\|^2_{\sigma,\mathcal{R}}=1. \tag{10}$$

*Proof.* Choose Bloch coordinates in which

$$P_1-P_0=sZ,\qquad \sigma=\frac{1}{2}(I+cX_0+bsZ),\qquad c=\sqrt{1-s^2},\quad b=2r-1.$$

Here $X_0$ denotes the Pauli matrix, not the signal difference $X$. Set

$$\ell^2=c^2+b^2s^2=1-4vs^2.$$

The eigenvalues of $\sigma$ are $(1\pm\ell)/2$ and

$$\det\sigma=vs^2. \tag{11}$$

In the eigenbasis of $\sigma$, the absolute diagonal and off-diagonal components of $X=sZ$ are, respectively,

$$\frac{|b|s^2}{\ell},\qquad \frac{sc}{\ell}.$$

Using (11), direct substitution gives

$$vR = \frac{b^2 s^2 + 2sc^2\sqrt{v}}{\ell^2}. \tag{12}$$

The difference between $s\ell^2$ and the numerator in (12) factors as

$$s\ell^2 - \left(b^2 s^2 + 2sc^2\sqrt{v}\right) = s(1-s)(1-2\sqrt{v})(1-2s\sqrt{v}) \geq 0,$$

which proves (9).

For (10), use the eigenbasis of $\sigma$. Since $\sigma$ is diagonal,

$$\langle \sigma, X \rangle_{\sigma,\mathcal{R}} = \sum_a \frac{\lambda_a X_{aa}}{\lambda_a} = \operatorname{Tr} X = 0,$$

while $\|\sigma\|^2_{\sigma,\mathcal{R}} = \sum_a \lambda_a = 1$. □

As checks on the constants, at $\alpha = 1/2$ the bound (5) reads $k_{1/2}(x,y) \leq 1/(4\sqrt{xy})$. When $s = 1$, (9) is equality: the channel is classical, the relevant perturbations commute with the reference, and the coefficient $\alpha(1-\alpha)$ agrees with the ordinary scalar Hessian.

## 4 Fourier orthogonality and proof of the theorem

For each coordinate define

$$\sigma_i = (1 - r_i)P_{i,0} + r_i P_{i,1}, \qquad X_i = P_{i,1} - P_{i,0}.$$

The channel sends the two elementary signed measures to

$$\Phi_i(q_i) = \sigma_i, \qquad \Phi_i(q_i F_i) = \sqrt{v_i}\, X_i.$$

It follows from (1) that

$$\sigma = \bigotimes_i \sigma_i, \qquad D := \rho'_t = \sum_T \widehat{f}(T) \bigotimes_{i \in T} (\sqrt{v_i} X_i) \otimes \bigotimes_{i \notin T} \sigma_i. \tag{13}$$

By (8) and (10), two different summands in (13) are orthogonal in the $\mathcal{R}$ metric: a coordinate in their symmetric difference contributes the factor $\langle \sigma_i, X_i \rangle_{\sigma_i,\mathcal{R}} = 0$. Lemma 3 therefore gives

$$\begin{aligned} \|D\|^2_{\sigma,\mathcal{R}} &= \sum_T \widehat{f}(T)^2 \prod_{i \in T} v_i R_i \\ &\leq \sum_T \widehat{f}(T)^2 \prod_{i \in T} s_i. \end{aligned} \tag{14}$$

Combining (14) with Lemma 2 proves the first inequality in (3).

It remains to calculate the erasure Hessian. Given a reveal set $S$,

$$p_{t,S} = q_S(1 + t f_S), \qquad f_S = \mathbb{E}_q[f \mid X_S].$$

Independence and centering of the $F_i$ imply

$$f_S = \sum_{T \subseteq S} \widehat{f}(T) F_T, \qquad \mathbb{E}_{q_S} f_S^2 = \sum_{T \subseteq S} \widehat{f}(T)^2.$$

Consequently,

$$\begin{aligned} -C''_\alpha(0) &= \alpha(1-\alpha)\mathbb{E}_S\mathbb{E}_{q_S} f_S^2 \\ &= \alpha(1-\alpha)\sum_T \widehat{f}(T)^2 \Pr[T \subseteq S] \\ &= \alpha(1-\alpha)\sum_T \widehat{f}(T)^2 \prod_{i\in T} s_i. \end{aligned}$$

This is the equality in (3). The zeroth- and first-derivative statements follow from normalization and $\mathbb{E}_q f = 0$, completing the proof of Theorem 1.

## 5 The local TC–Heart consequence

**Corollary 4** (Local TC–Heart at every product law)**.** *Assume (2), so that $q$ is the product of the one-bit marginals of every $p_t$. Then*

$$\left.\frac{d^2}{dt^2} D(\rho_t\|\sigma)\right|_{t=0} \le \left.\frac{d^2}{dt^2}\mathbb{E}_S D(p_{t,S}\|q_S)\right|_{t=0} = \sum_{|T|\ge 2} \widehat{f}(T)^2 \prod_{i\in T} s_i. \tag{15}$$

*Proof.* In finite dimension, for full-rank states,

$$\operatorname{Tr}\rho^\alpha\sigma^{1-\alpha} = 1 - (1-\alpha)D(\rho\|\sigma) + O((1-\alpha)^2) \qquad (\alpha \uparrow 1),$$

and the analogous scalar expansion holds for every marginal. Divide (3) by $1-\alpha$ and let $\alpha \uparrow 1$; analyticity permits interchange with the two $t$ derivatives. Since $\alpha(1-\alpha)/(1-\alpha) \to 1$, we obtain (15). □

## 6 A global endpoint for the next curvature lemma

The local proof above suggests comparing the full midpoint curvature along a ray. A resolvent representation reduces that proposed comparison to a family indexed by a cutoff $L \ge 0$. Although the finite-$L$ statement remains open, its $L \to \infty$ leading term admits a simple global proof in every dimension.

Let

$$G = \bigotimes_i \begin{pmatrix} 1 & \sqrt{1-s_i^2} \\ \sqrt{1-s_i^2} & 1 \end{pmatrix}, \qquad B = D_{\sqrt{q}} G D_{\sqrt{q}} = \bigotimes_i B_i.$$

For a real function $f$ on the cube, let $D_f$ denote its diagonal matrix and put $H = B^{1/2} D_f B^{1/2}$. As before, $R$ is a random reveal set with $\Pr[i \in R] = s_i$, independently.

**Theorem 5** (Global large-cutoff coefficient)**.** *For every real $f$, without positivity or marginal restrictions,*

$$\operatorname{Tr}(H^2 B^{1/2}) \le \mathbb{E}_R \mathbb{E}_q\left[\mathbb{E}_q[f \mid X_R]^2\right]. \tag{16}$$

*Proof.* Both sides of (16) are quadratic forms in the vector of values of $f$. Cyclicity of trace gives

$$\operatorname{Tr}(H^2 B^{1/2}) = \operatorname{Tr}(D_f B D_f B^{3/2}) = f^{\mathsf{T}}\left(B \circ B^{3/2}\right) f.$$

On the erasure side, conditioning one coordinate either averages it or reveals it. Independence of the reveal decisions therefore gives the quadratic-form matrix

$$K_E = \bigotimes_i \left( (1 - s_i) q_i q_i^{\mathsf{T}} + s_i D_{q_i} \right).$$

It remains to compare the two matrices one coordinate at a time. Write $q_i = (q_0, q_1)^{\mathsf{T}}$ and $v = q_0 q_1$. Since $B_i$ is a density matrix, $0 \le B_i \le I$, hence $B_i^{3/2} \le B_i$. The Schur product theorem gives

$$B_i \circ B_i^{3/2} \le B_i \circ B_i.$$

Moreover, a direct calculation using $(B_i)_{01}^2 = (1 - s_i^2)v$ shows

$$(1 - s_i) q_i q_i^{\mathsf{T}} + s_i D_{q_i} - B_i \circ B_i = s_i v \begin{pmatrix} 1 & -(1 - s_i) \\ -(1 - s_i) & 1 \end{pmatrix} \ge 0.$$

Thus $B_i \circ B_i^{3/2} \le (1 - s_i) q_i q_i^{\mathsf{T}} + s_i D_{q_i}$. Tensor products preserve this order (by a telescoping expansion), proving (16). □

In the resolvent formulation of midpoint ray convexity, the quantum and erasure integrands have the respective asymptotics

$$J_Q(\lambda) = \lambda^{-3} \operatorname{Tr}(H^2 B^{1/2}) + O(\lambda^{-4}), \qquad J_E(\lambda) = \lambda^{-3} \mathbb{E}_R \mathbb{E}_q \big[ \mathbb{E}_q[f \mid X_R]^2 \big] + O(\lambda^{-4}).$$

Theorem 5 therefore proves the large-cutoff endpoint of that global comparison. It does not control finite cutoffs.

**Remark 6** (What remains global)**.** *The finite-displacement conjecture would assert*

$$\operatorname{Tr} \rho_p^{\alpha} \rho_q^{1-\alpha} \ge \mathbb{E}_S \sum_z p_S(z)^{\alpha} q_S(z)^{1-\alpha}, \qquad q = \bigotimes_i p_i,$$

*at least for $1/2 \le \alpha < 1$. Theorem 1 proves its complete quadratic expansion around $p = q$, in fact for all $0 < \alpha < 1$. A global proof still requires controlling the Petz Hessian away from the product point, where the reference remains product but the moving state no longer tensorizes and the exact Fourier orthogonality used in (14) is lost.*